Ex no:1	Mastering Git & GitHub Command-Line Essentials
Date	

Aim:

To learn understand and practice the basic Git commands for repository creation, version control, branch management, and remote repository operations.

Git & GitHub Commands:

1. git init

Description: Creates a local copy of an existing remote Git repository.

Syntax: git init
Output:

```
nevesaxxd@Nevesas-MacBook-Air gitlub-tutorial %
git init
Reinitialized existing Git repository in /Users/nevesaaxd
/Documents/github-tutorial/.git/

nevesaxxd@Nevesas-MacBook-Air github-tutorial %
```

2. git clone

Description: Creates a local copy of an existing remote Git repository.

Syntax: git clone https://github.com/antonyrajinevesa-del/sample

Output:

```
nevesaaxd@Nevesas-MacBook-Air github-tutorial %
git clone https://github.com/sanjitha-07/sample
Cloning into 'sample'...
remote: Enumrrating objets: 24, done.
remote: Counting objects: 100% (24/24), done.
remote: Compressing objects: 100% (17/17), done.
remote: Total 24 (delta 4), reused 17 (delta 2), pused 0 (from 0)
Receiving objects: 100% (4/4), done.
nevesaaxd@Nevesas-MacBook-Air github-tutorial %
```

3.git status

Description: Displays the current status of the working directory and staged changes.

Syntax: git status

```
nevesaaxd@Nevesas-MacBook-Air github-tutorial % git status
On branch b2
Your branch is up to date with 'origin/b2'.
Untracked files:
   (use 'git add <file>...' to include in what will be committed
       sample/
nothing added to commit but untracked files present (use
'git add' to track)
nevesaaxd@Nevesas-MacBook-Air github-tutorial %
```

4.git add

Description:

Stages files for the next commit.

Syntax: git add file_name

```
nevesaaxd@Nevesas-MacBook-Air ~ % cd Documents/github\ tutorial
nevesaaxd@Nevesas-MacBook-Air github\ tutorial % git add Todo.txt

nevesaaxd@Nevesas-MacBook-Air github\ tutorial %
```

5.git commit:

Description: Saves the staged changes to the local repository with a commit message.

Syntax: git commit -m "Commit message"

```
nevesaaxd@Nevesas-MacBook-Air github-tutorial % git
commit -m "Initial commit
[b2 241a0c9] Initial commit
1 file changed, 1 insertion(+)
create mode 160000 sample
nevesaaxd@Nevesas-MacBook-Air github-tutorial %
```

6.git push

Description:

Uploads local commits to a remote repository such as GitHub.

Syntax:
git push <remote_name> <branch_name>

```
nevesaaxd@Nevesas-MacBook-Air github-tutorial % git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 thrads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 378 bytes | 378.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/sanjitha-07/sample.git
   7ee092f..082f299  main -> main
nevesaaxd@Nevesas-MacBook-Air github-tutorial %
```

7.gitcheckout:

Description: Switches from one branch to another**.**

Syntax: git checkout <branch_name>

```
nevesaaxd@Nevesas-MacBook-Air github-tutorial % git checkout
Your branch is up to date with 'origin/main'.

nevesaaxd@Nevesas-MacBook-Air ~ %
```

8.gitcheckout-b:

Description: Creates a new branch and switches to it immediately.

Syntax: git checkout -b <new_branch_name>

```
nevesaaxd@Nevesas-MacBook-Air ~ % cd Documets/github tutoria1
nevesaaxd@Nevesas-MacBook-Air github\ tutoria1
nevesaaxd@Nevesas-MacBook-Air % git git checkout -b feature1
Switched to a new branch 'feature1'
nevesaaxd@Nevesas-MacBook-Air github\tutoria1 %
```

9.gitbranch:

Description: Lists, creates, or deletes Git branches.

Syntax: git branch (list branches)

```
nevesaaxd@Nevesas-MacBook-Air github\ tutorial %
git branch
   b1
   b2
   * feature1
   main
nevesaaxd@Nevesas-MacBook-Air github\ tutorial %
```

10.gitconfig:

Description: Configures Git settings such as username, email, and other preferences.

Syntax: git config --global user.name "<Your Name>"

```
nevesaaxd@Nevesas-MacBook-Air github\ tutorial % git config -global user.name"'antonyrajivesade"-
```

GitHub commands syntax & Description (10)	Implementation and Execution (15)	Tool/ Technology usage (10)	Code Quality & Documentation (5)	Time Management (5)	Viva (5)	Total (50)

Result:

Successfully performed basic Git operations, including repository creation, version control, branch management, and pushing changes to GitHub.